YOLO Environment Setup

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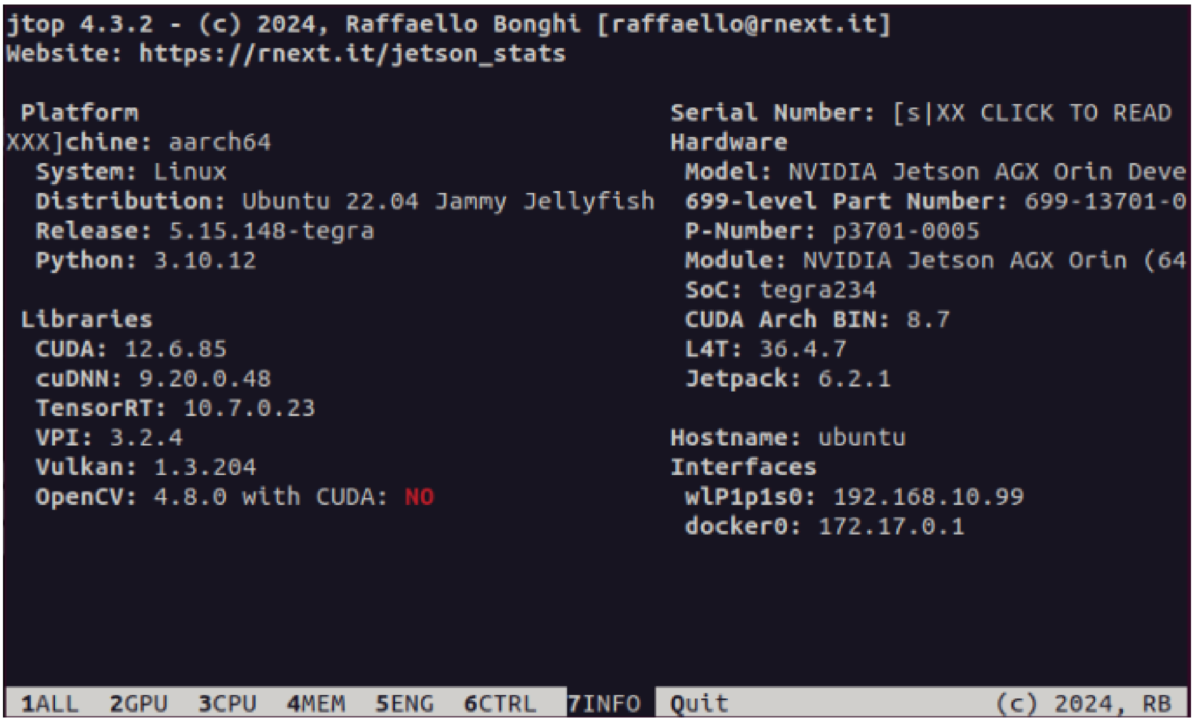
1、System Information



```
jtop 4.3.2 - (c) 2024, Raffaello Bonghi [raffaello@rnext.it]
Website: https://rnext.it/jetson_stats

 Platform                                    Serial Number: [s|XX CLICK TO READ
XXX]chine: aarch64                           Hardware
  System: Linux                               Model: NVIDIA Jetson AGX Orin Deve
  Distribution: Ubuntu 22.04 Jammy Jellyfish  699-level Part Number: 699-13701-0
  Release: 5.15.148-tegra                     P-Number: p3701-0005
  Python: 3.10.12                             Module: NVIDIA Jetson AGX Orin (64
                                              SoC: tegra234
 Libraries                                    CUDA Arch BIN: 8.7
  CUDA: 12.6.85                               L4T: 36.4.7
  cuDNN: 9.20.0.48                            Jetpack: 6.2.1
  TensorRT: 10.7.0.23
  VPI: 3.2.4                                 Hostname: ubuntu
  Vulkan: 1.3.204                            Interfaces
  OpenCV: 4.8.0 with CUDA: NO                 wlP1p1s0: 192.168.10.99
                                              docker0: 172.17.0.1

 1ALL  2GPU  3CPU  4MEM  5ENG  6CTRL 7INFO Quit                  (c) 2024, RB
```

2、Download Files

Note: If you have already set up the YOLO11 environment, you need to rename the ultralytics folder in the user directory first, then download the files.

Here we use ultralytics-v8.4.21 as an example

```
cd ~/
git clone https://github.com/ultralytics/ultralytics.git -b v8.4.21 --depth=1
```

```
jetson@ubuntu:~$ git clone https://github.com/ultralytics/ultralytics.git -b v8.
4.21 --depth=1
Cloning into 'ultralytics'...
remote: Enumerating objects: 979, done.
remote: Counting objects: 100% (979/979), done.
remote: Compressing objects: 100% (827/827), done.
remote: Total 979 (delta 209), reused 290 (delta 137), pack-reused 0 (from 0)
Receiving objects: 100% (979/979), 2.44 MiB | 59.00 KiB/s, done.
Resolving deltas: 100% (209/209), done.
Note: switching to '83471a5041fe5d7fdba5c12aff65f4f84f27c081'.

You are in 'detached HEAD' state. You can look around, make experimental
changes and commit them, and you can discard any commits you make in this
state without impacting any branches by switching back to a branch.
```

If you cannot clone the repository due to network issues, you can download the compressed file from the following URL and extract it.

```
https://github.com/ultralytics/ultralytics/releases/tag/v8.4.21
```

3、Install Ultralytics

```
cd ~/ultralytics
sudo pip3 install ".[export]" -i https://pypi.tuna.tsinghua.edu.cn/simple
```

```
jetson@ubuntu:~/ultralytics$ sudo pip3 install ".[export]" -i https://pypi.tuna.
tsinghua.edu.cn/simple
Looking in indexes: https://pypi.tuna.tsinghua.edu.cn/simple
Processing /home/jetson/ultralytics
  Installing build dependencies ... done
  Getting requirements to build wheel ... done
  Preparing metadata (pyproject.toml) ... done
Requirement already satisfied: numpy>=1.23.0 in /usr/local/lib/python3.10/dist-p
ackages (from ultralytics==8.4.21) (1.26.4)
Requirement already satisfied: matplotlib>=3.3.0 in /usr/local/lib/python3.10/di
st-packages (from ultralytics==8.4.21) (3.10.8)
Requirement already satisfied: opencv-python>=4.6.0 in /usr/local/lib/python3.10
/dist-packages (from ultralytics==8.4.21) (4.11.0.86)
Requirement already satisfied: pillow>=7.1.2 in /usr/lib/python3/dist-packages (
from ultralytics==8.4.21) (9.0.1)
```

5、Configure GPU Acceleration

Since we have already installed torch, torchvision and onnxruntime-gpu in the previous tutorial, there is no need to run the installation commands here.

> torch

> torchvision

> onnxruntime-gpu

6、Verify Installation

> Verify Ultralytics

```
python3 -c "import ultralytics; print(ultralytics.__version__)"
```

> **Verify Torch**

```
python3 -c "import torch; print(torch.__version__);
print(torch.cuda.is_available())"
```

> **Verify Torchvision**

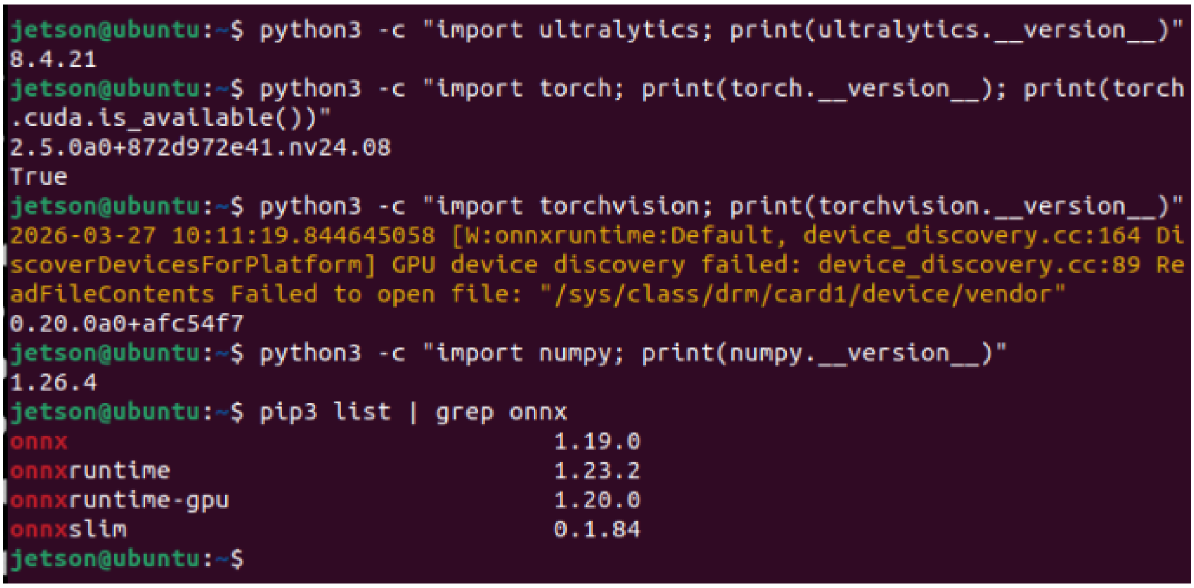
```
python3 -c "import torchvision; print(torchvision.__version__)"
```

> **Verify Numpy**

```
python3 -c "import numpy; print(numpy.__version__)"
```

> **Verify onnxruntime-gpu**

```
pip3 list | grep onnx
```



References

https://docs.ultralytics.com/guides/nvidia-jetson/